



Service Bulletin Number: 6508353

Released Date: 23-oct-2023

Water In the Fuel System and Corrosion on Fuel System Components

## Water In the Fuel System and Corrosion on Fuel System Components

- The defects outlined in this bulletin are **not** material or factory workmanship related.
- Repairs related to the issues outlined below are **not** warrantable and should **not** be submitted under warranty.
- Beginning in May 2022, Cummins Inc. Warranty Quality will begin flagging a limited number of claims to incorporate component analysis into the claim adjudication decision.
  - The component analysis will be related to certain parts returned as part of the EPR (Enhanced Part Return) process, that are used for failure analysis today. The intention of the changes is to leverage failure analysis to improve service information such as diagnostic content and repair procedures, as well as for identifying improvement opportunities related to repair practices.
  - If the component analysis result supports the failure and replacement of the component, claims will be processed as they are today, within Cummins Warranty Quality guidelines.
  - If the analysis concludes that the component was replaced contradictory to published service information or due to **not** being a defect in Cummins® material or factory workmanship, or if the return shipment of the part is delayed, the claim may be subject to adjustment or denial.

### Product Affected

- The contents of this Service Bulletin apply exclusively to the Bosch® fuel system.

### Issue

Water in the fuel systems will cause corrosion on fuel system components. This document outlines ways to identify corrosion on fuel system components and detect water in the fuel.

#### Symptom:

- Corrosion on fuel system components

#### Root Cause

- Water in the fuel system causes damage to the fuel system.

### Verification and Resolution

- Inspect fuel system components for corrosion.
  - a. **Only** if directed by published troubleshooting to remove the specific parts listed below, check parts listed below for corrosion. Do **not** remove parts outside of those directed by published troubleshooting.
  - b. Parts to inspect for corrosion if directed to remove according to published troubleshooting:
    - i. Injector fuel supply connector (For example of corrosion, see Figure 2)
    - ii. Fuel Injector
    - iii. Fuel pump electronic fuel control actuator (For example of corrosion, see Figure 4)
    - iv. Fuel rail pressure relief valve (For example of corrosion, see Figure 6)
    - v. Fuel pump pressure regulator. (For example of corrosion, see Figures 12 and 13)
- **Only** If corrosion is found on any of the components listed above, proceed to inspect the fuel tank and see Service Bulletin, Water in Fuel Contamination, Bulletin 5659823 for restoring the fuel system.
- Sample the fuel and inspect for signs of water in the fuel tank.
  - a. Sample fuel from the lowest point of the fuel tank.
  - b. Siphon fuel or use fuel tank drain plug and drain fuel into a clear container.

- c. Let the fuel sit for 5 minutes.
- Check fuel sample for signs of water in the clear container.
  - a. If free-standing water is found in the bottom of the clear container (See Figure 7), see Service Bulletin, Water in Fuel Contamination, Bulletin 5659823 for repair.
  - b. If no free-standing water is found at the bottom of the container:
    - i. Print fuel bar chart. Select hyperlink below for fuel bar chart.

**Note :**

[https://quickserve.cummins.com/protected/files/qsol/cms/distillate\\_fuel\\_bar\\_chart.pdf](https://quickserve.cummins.com/protected/files/qsol/cms/distillate_fuel_bar_chart.pdf)  
([https://quickserve.cummins.com/protected/files/qsol/cms/distillate\\_fuel\\_bar\\_chart.pdf](https://quickserve.cummins.com/protected/files/qsol/cms/distillate_fuel_bar_chart.pdf))

- ii. Place fuel bar chart behind the fuel sample and compare with examples in Figure 8-10.
- iii. If Line 1 on fuel bar chart is **not** visible while looking through fuel sample, see Service Bulletin, Water in Fuel Contamination, Bulletin 5659823 for restoring the fuel system.

**Examples of Corrosion:**

Signs of corrosion can be seen in the fuel pump electronic fuel control actuator, fuel rail pressure relief valve, and the injector fuel supply connector. See Figures below for examples.

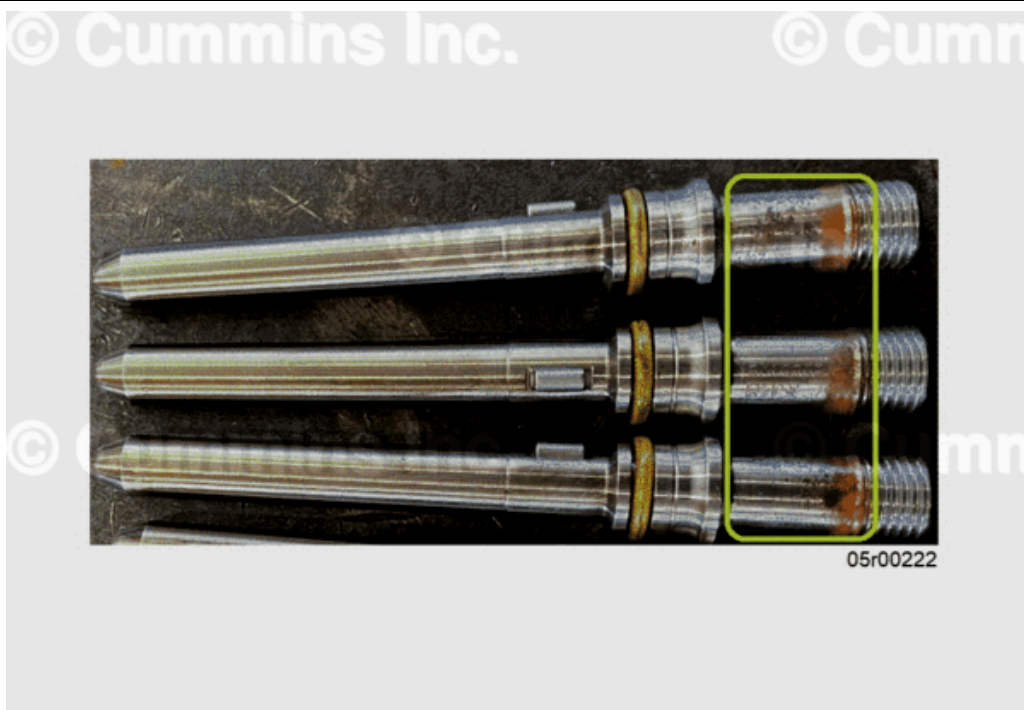


Figure 1, Normal and Acceptable Corrosion on Injector Connector Above O ring



Figure 2, Abnormal and Unacceptable Corrosion on Injector Connector Below O ring

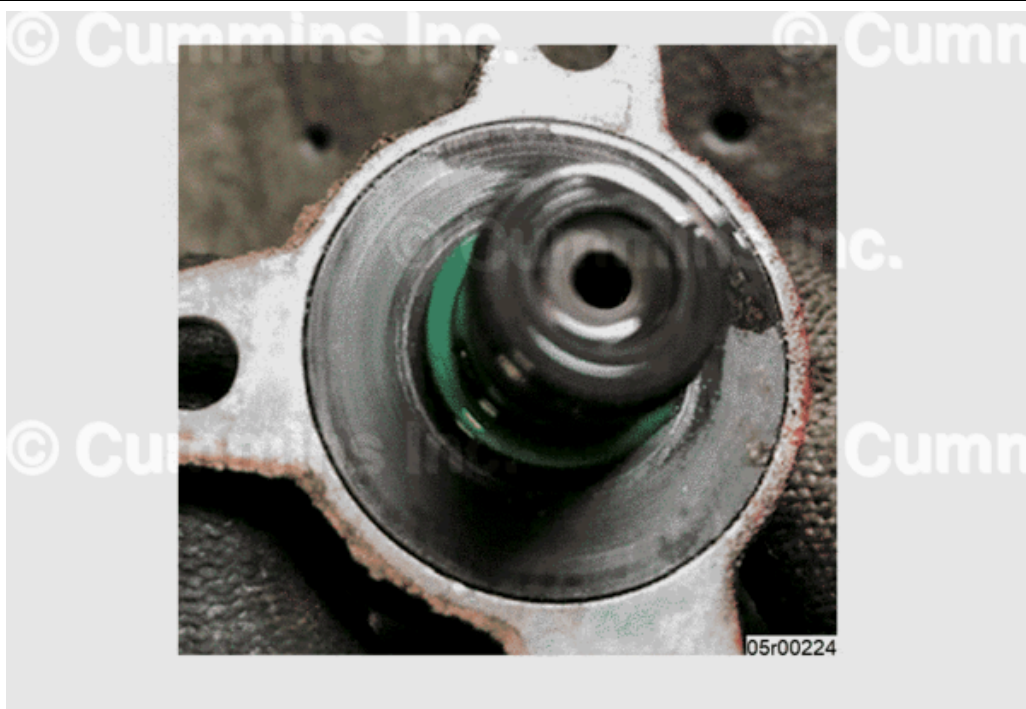


Figure 3, Clean and Acceptable Fuel Pump Electronic Fuel Control Actuator



Figure 4, Corroded and Unacceptable Fuel Pump Electronic Fuel Control Actuator



Figure 5, Clean and Acceptable Fuel Rail Pressure Relief Valve Corrosion





Figure 6, Corroded and Unacceptable Fuel Rail Pressure Relief Valve Corrosion



Figure 7, Free Standing Water in Diesel Fuel.



Figure 8, 0 ml of Water Added to Diesel Fuel



Figure 9, 1 ml [ .0 fl-oz ] Water Added to Diesel Fuel

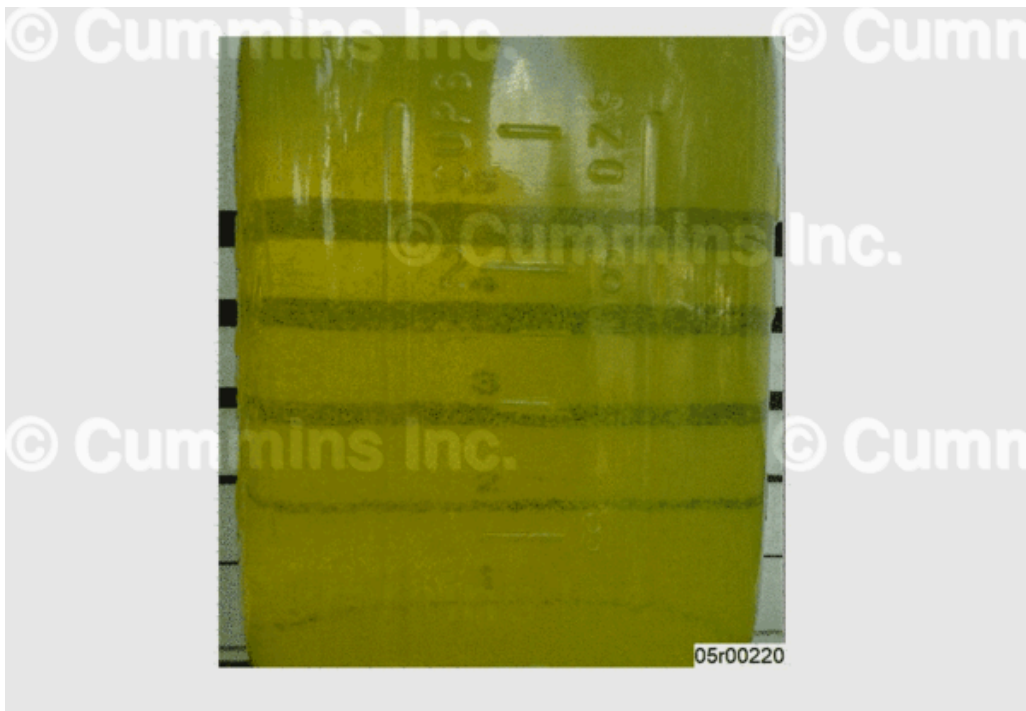


Figure 10, 3 ml [ .10 fl-oz ] Water Added to Diesel Fuel



Figure 11, 5 ml [ .17 fl-oz ] Water Added to Diesel Fuel



Figure 12, Example of Corrosion in Fuel Pump Pressure Regulator



Figure 13, Example of Corrosion in Fuel Pump Pressure Regulator





Figure 14, Fuel Pump Pressure Regulator

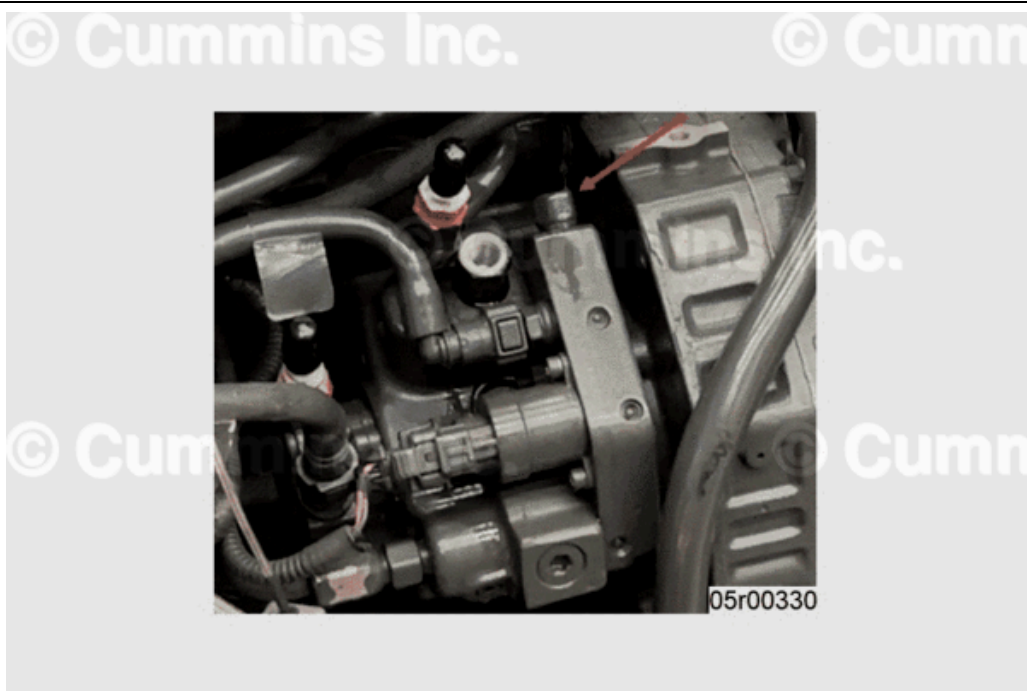


Figure 15, Location of Fuel Pump Pressure Regulator



Figure 16, Example of Corrosion on Injector

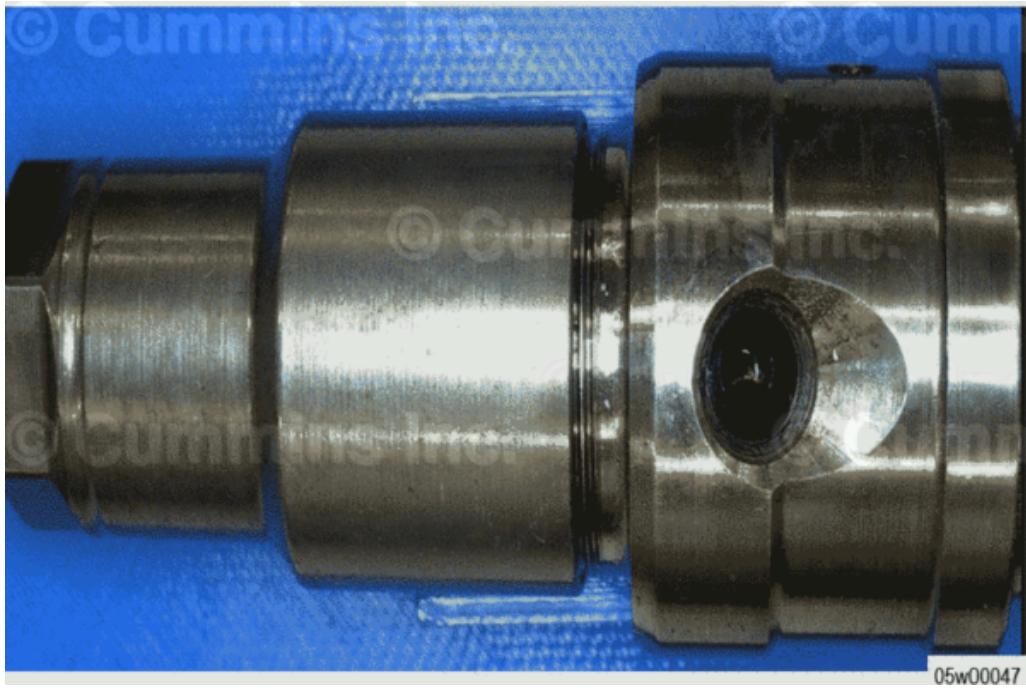


Figure 17, Example of Non-Corroded Injector

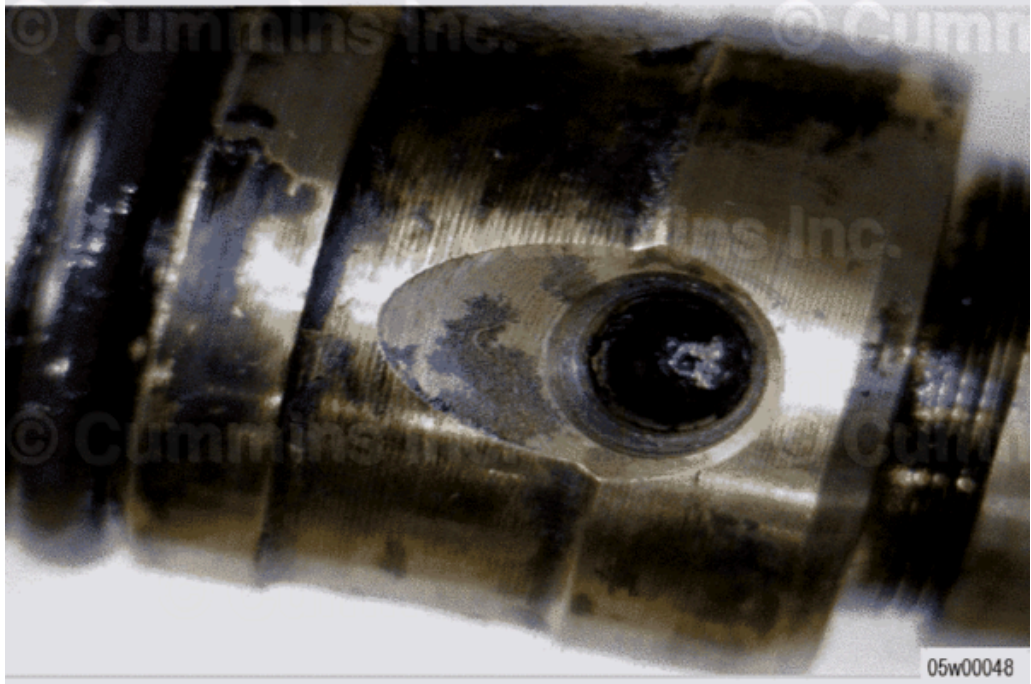


Figure 18, Example of Corrosion on Injector



Figure 19, Example of Corrosion on Injector

Document History

| Date       | Details                                         |
|------------|-------------------------------------------------|
| 2023-10-23 | Module Created                                  |
| 2024-7-2   | none                                            |
| 2024-11-15 | Added Figures 17, 18, and 19                    |
| 2025-1-14  | Added B181B and B182B to Product Affected list. |

| Date      | Details                                       |
|-----------|-----------------------------------------------|
| 2025-1-24 | Updated Product Affected.                     |
| 2025-2-12 | Updated Product Affected and removed Table 1. |

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Last Modified: 17-Feb-2025

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